IRXBot — Wiring Pinout

Jetson Orin Nano · STM32 NUCLEO-F446RE · ESP32-S3 DevKitC-1 | Taken from the firmware on `main`, 2026-11-02

Voltage colours: **3.3 V** · **5 V** · **12 V** · **24 V** · **GND**.

Before power-on: (1) All boards share one common GND. (2) STM32 and ESP32 signal pins are **3.3 V**; never put 5 V, 12 V or 24 V on a signal pin. (3) CAN: about **60 Ω** between CAN-H and CAN-L with power off. (4) The E-stop must cut the contactor — test it last.

1. Jetson Orin Nano

Power and network

Connector	Goes to	Voltage	Notes
DC jack	Own 24 V → 19 V DC-DC, through the Jetson branch fuse	**19 V**	Not shared with other loads
Ethernet	Network	—	Wi-Fi is built in

USB — now (temporary, until the second Waveshare hub arrives)

Port	Device	USB ID	Notes
Jetson USB 1	RealSense D435i	`8086:0b3a`	Direct to the Jetson, USB 3 cable
Jetson USB 2	STM32 Nucleo (ST-LINK USB)	—	`/dev/ttyACM*`, 115200. Also powers the Nucleo
Jetson USB 3	ESP32-S3 port marked **USB** (native)	`303a:1001`	Telemetry + IP for the LCD. Leave the other ESP32 port (UART) and its 5V pin empty
Jetson USB 4	Waveshare hub 1 (powered)	—	Check the input voltage on its label before powering
Waveshare 1	4G modem EC25-EUX	`2c7c:0125`	Antenna on MAIN
Waveshare 2	RPLIDAR C1 (its USB adapter)	CP210x	460800 baud, 5 V from USB
Waveshare 3	360° camera	UVC	
Waveshare 4	Black 4-port hub (unpowered)	—	Strap it down, all port switches ON
Black hub 1	ReSpeaker mic array	`2886:0018`	3.5 mm out → amplifier
Black hub 2	Thermal camera USB-serial adapter (FT232)	FTDI	Adapter jumper at 3.3 V

USB — final (two Waveshare hubs)

Jetson	Waveshare hub 1	Waveshare hub 2
RealSense · STM32 · ESP32 · Hub 1	ReSpeaker · 360° camera · LiDAR · Hub 2	Thermal adapter · 4G modem

Thermal camera (MLX90640 board) → USB-serial adapter

Thermal board	Adapter	Voltage	Notes
VIN	3V3	**3.3 V**	Use 5V only if the adapter has no 3.3 V pin
GND	GND	**GND**	
TX	RX	**3.3 V**	Crossed
RX	TX	**3.3 V**	Crossed
SCL, SDA, NC, PS	—	—	Leave unconnected. Do not bridge the SET I2C pads

Audio

From	To	Voltage	Notes

12 V rail → 5 A fuse	XH-M139 amplifier DC + / −	**12 V**	Never the raw battery bus (reaches ~29 V)
ReSpeaker 3.5 mm	Amplifier audio in (L, R, GND)	Audio	
Amp L + / L −	Left speaker + / −	Audio	Speaker − never to GND
Amp R + / R −	Right speaker + / −	Audio	Subwoofer output left empty

2. STM32 NUCLEO-F446RE

> "Header" = the Arduino-style header name printed on the Nucleo. Pins with no header name are on the long morpho headers (CN7 / CN10). **Do not wire PA2 / PA3 (D1 / D0):** they carry the USB link to the Jetson. The ToF pin labels in CubeMX (TOF_XSHUT_FRONT etc.) are out of date; this table follows the firmware.

Links

Pin	Header	Goes to	Voltage	Notes
USB (ST-LINK)	CN1	Jetson USB	**5 V**	USART2, 115200 — Jetson commands
PC10	—	ESP32 GPIO 35 (RX)	**3.3 V**	USART3 TX → ESP32, 115200
PC5	—	ESP32 GPIO 36 (TX)	**3.3 V**	USART3 RX ← ESP32
GND	GND	ESP32 GND	**GND**	Needed for the UART link

CAN → SN65HVD230 transceiver → 4 × VESC

Pin	Header	Goes to	Voltage	Notes
PA12	—	SN65HVD230 CTX / TX	**3.3 V**	CAN1 TX, 500 kbps
PA11	—	SN65HVD230 CRX / RX	**3.3 V**	CAN1 RX
3V3	3V3	SN65HVD230 VCC	**3.3 V**	3.3 V part — not 5 V
GND	GND	SN65HVD230 GND	**GND**	
CAN-H / CAN-L / GND	—	VESC 1 FR · 2 FL · 3 RR · 4 RL	Bus	3 wires, twisted H/L. 120 Ω only at the two ends of the bus

Sensors

Pin	Header	Goes to	Voltage	Notes
PB8	D15	All 5 ToF SCL	**3.3 V**	I2C1
PB9	D14	All 5 ToF SDA	**3.3 V**	I2C1
PB0	A3	ToF **RIGHT** XSHUT	**3.3 V**	Becomes address 0x30
PA8	D7	ToF **LEFT** XSHUT	**3.3 V**	0x31
PC7	D9	ToF **FRONT LEFT** XSHUT	**3.3 V**	0x32
PB6	D10	ToF **FRONT RIGHT** XSHUT	**3.3 V**	0x33
PC9	—	ToF **REAR** XSHUT	**3.3 V**	0x34
3V3 / GND	3V3 / GND	All 5 ToF VIN / GND	**3.3 V**	
PB10	D6	BNO085 IMU SCL	**3.3 V**	I2C2 — separate bus from ToF
PC12	—	BNO085 IMU SDA	**3.3 V**	I2C2
PC8	—	BNO085 RST	**3.3 V**	Pulsed low at boot

3V3 / GND	3V3 / GND	BNO085 VIN / GND	**3.3 V**	

Bumpers, RC, horn, contactor

Pin	Header	Goes to	Voltage	Notes
PC0	A5	Bumper front left	**3.3 V**	Switch between pin and GND (internal pull-up). Firmware reads **pin HIGH = pressed**, so use the switch's NC contact (opens when hit)
PC1	A4	Bumper front right	**3.3 V**	
PC2	—	Bumper rear left	**3.3 V**	
PC3	—	Bumper rear right	**3.3 V**	
PA10	D2	FS-iA10B **iBUS** servo signal	**signal**	USART1 RX. Use the iBUS port, not PWM or SBUS
5 V rail / GND	—	FS-iA10B power / GND	**5 V**	Regulated 5 V; GND common with STM32
PA9	D8	Not used	—	USART1 TX
PC6	—	Horn MOSFET/relay TRIG	**3.3 V**	Horn itself on **12 V**, own fuse
PB12	—	XY-MOS board TRIG (contactor)	**3.3 V**	HIGH = contactor on. See section 4

3. ESP32-S3 DevKitC-1 (N16R8)

> **Do not use:** GPIO 19 / 20 (native USB), 26–32 (flash), 33–37 (PSRAM). GPIO 46 must be LOW at boot. Power: from the Jetson through the **USB** port; do not also feed the 5V pin.

Links

GPIO	Goes to	Voltage	Notes
USB port	Jetson USB	**5 V**	Native USB (GPIO 19/20), `303a:1001`
35	STM32 PC10	**3.3 V**	UART1 RX
36	STM32 PC5	**3.3 V**	UART1 TX

I2C sensors — all on GPIO 8 (SDA) and GPIO 9 (SCL)

Sensor	Address	Voltage	Notes
INA228	0x40	**3.3 V**	2 mΩ shunt in the 24 V bus; its VBUS pin reads the **24 V** bus
BME280	0x76	**3.3 V**	Firmware expects 0x76
SGP30	0x58	**3.3 V**	
SPS30	0x69	**5 V**	Needs 5 V. **SEL pin → GND** for I2C. SDA/SCL still on GPIO 8/9
VEML7700	0x10	**3.3 V**	Had failed reads — check its wires

Other sensors

GPIO	Goes to	Voltage	Notes
4	Both DS18B20 DATA (battery + DC-DC)	**3.3 V**	Same wire; 4.7 kΩ pull-up DATA → 3.3 V
1	MQ-2 AO	**5 V**	Module on 5 V, but GPIO 1 takes 3.3 V max — divide AO down if it can exceed 3.3 V
2	12 V rail divider (56k / 10k, 100 nF)	**12 V**	Divider top to 12 V, middle to GPIO

6	5 V rail divider (22k / 10k, 100 nF)	**5 V**	
7	3.3 V rail divider (10k / 10k, 100 nF)	**3.3 V**	

LCD ST7789 (SPI)

GPIO	LCD pin	Voltage	Notes
10	CS	**3.3 V**	
11	MOSI / SDA	**3.3 V**	
12	SCK / SCL	**3.3 V**	
41	DC	**3.3 V**	
5	RST	**3.3 V**	
3V3 / GND	VCC (and BL) / GND	**3.3 V**	No GPIO controls the backlight

Lights and buzzer — GPIO drives a MOSFET; the load runs from its own rail

GPIO	Goes to	Load voltage	Notes
38	WS2812B LEFT strip DIN (23 LEDs)	**5 V**	Through level shifter + 330 Ω. Strip power from the 5 V rail, not the ESP32 board
39	WS2812B RIGHT strip DIN (23 LEDs)	**5 V**	Same as left; bulk capacitor at the strip
15	Front amber LEFT MOSFET	**12 V**	On/off
16	Front amber RIGHT MOSFET	**12 V**	On/off
17	Rear red LEFT MOSFET	**12 V**	PWM (tail / brake)
18	Rear red RIGHT MOSFET	**12 V**	PWM
14	Spotlight MOSFET	**12 V**	Own branch fuse
21	Buzzer MOSFET	**12 V**	Active HIGH, own fuse

4. Power and safety

Rail	Source	Voltage	Feeds
Battery bus	2 × 12.8 V $LiFePO_4$ in series → XT90 → main fuse → contactor	**25.6 V (up to ~29 V)**	4 × VESC, each with its own fuse
24 V	Mean Well DDR-120B-24, fused	**24 V**	Electronics, contactor coil chain — not the VESCs
19 V	Own DC-DC, Jetson fuse	**19 V**	Jetson only
12 V	Mean Well DDR-120B-12, fused	**12 V**	Lights, horn, buzzer, amplifier (5 A fuse)
5 V	24 → 5 V converter, fused	**5 V**	WS2812B strips, RC receiver, SPS30, MQ-2
3.3 V	Nucleo / ESP32 3V3 pins	**3.3 V**	Small sensors only (I2C, ToF, IMU, CAN transceiver, LCD)

Contactor chain (no software in the path)

From	To	Voltage

`24 V control`	Rocker / key switch → E-stop **NC** → XY-MOS VIN+	**24 V**
`XY-MOS VIN− and GND`	GND	**GND**
`XY-MOS TRIG`	STM32 PB12	**3.3 V**
`XY-MOS OUT+`	Contactor coil A1 (+)	**24 V**
`Coil A2 (−)`	GND directly	**GND**
`Flyback diode`	Stripe (cathode) → A1, plain end (anode) → A2	—

After each section

Section	Check
Power	Rails read about 24 / 12 / 5 / 3.3 V before plugging boards in
CAN	~60 Ω H to L with power off; `robot_check.py` hears all 4 VESCs
STM32 sensors	`robot_check.py --manual`: ToF, IMU, bumpers, RC
ESP32 sensors	`robot_check.py`: all ESP32 sensors OK
USB	`robot_check.py` finds every USB device
Last	Press the E-stop: the contactor must drop